

QMEA (Hypothesis Testing)

1. What is the main purpose of hypothesis testing?

- A. To prove that a hypothesis is definitely true
- B. To use sample data to evaluate a claim about a population
- C. To calculate the population mean exactly
- D. To eliminate sampling variability

2. A company introduces a new employee training program and tests whether it improves worker productivity.

The result is not statistically significant.

What should management conclude?

- A. The training program definitely does not work
- B. The training program definitely works
- C. There is insufficient evidence to conclude that the program improves productivity
- D. The company should conclude that productivity decreased

3. A company is considering launching a new product nationwide after a test-market study.

Suppose:

H_0 : The new product will not generate sufficient demand

What would be a Type I error?

- A. Launching a profitable product
- B. Not launching an unprofitable product
- C. Launching the product when demand is actually insufficient
- D. Not launching the product when demand would actually be sufficient

4. Using the same hypotheses:

H_0 : The new product will not generate sufficient demand

What would be a Type II error?

- A. Launching a product that fails
- B. Not launching a product that would actually have sufficient demand
- C. Launching a successful product
- D. Correctly rejecting an unsuccessful product

5. A company tests a new production technology. The statistical analysis shows strong evidence that the technology increases productivity.

What should management do next?

- A. Adopt the technology immediately because statistical significance guarantees profitability
- B. Ignore the statistical results
- C. Consider both the statistical evidence and the costs and benefits of implementation
- D. Conclude that productivity will increase equally for every factory

6. An online retailer tests whether a redesigned website increases the conversion rate. The study uses a very small sample and finds no statistically significant improvement.

What should management be cautious about concluding?

- A. The redesign definitely has no effect
- B. The redesign definitely decreases conversion
- C. The study may have insufficient power to detect a real improvement
- D. The null hypothesis must be false

7. A company introduces a performance bonus and wants to determine whether it increases employee productivity. Which alternative hypothesis best represents the management question?

- A. Productivity after the bonus program is lower
- B. Productivity after the bonus program is different
- C. Productivity after the bonus program is higher
- D. Productivity remains unchanged

8. A company claims that its batteries last an average of 10 hours. A researcher believes that the true average is less than 10 hours. Which hypotheses are appropriate?

- A. $H_0: \mu = 10, H_1: \mu < 10$
- B. $H_0: \mu < 10, H_1: \mu = 10$
- C. $H_0: \bar{X} = 10, H_1: \bar{X} < 10$
- D. $H_0: \mu = 10, H_1: \mu > 10$

9. At the beginning of a hypothesis test, which hypothesis is assumed to be true?

- A. The alternative hypothesis
- B. The research hypothesis
- C. The null hypothesis
- D. Both hypotheses

10. What is the primary goal of hypothesis testing?

- A. To prove that H_0 is true
- B. To prove that H_1 is true with certainty
- C. To determine whether there is enough evidence to support H_1
- D. To eliminate the possibility of making an incorrect decision